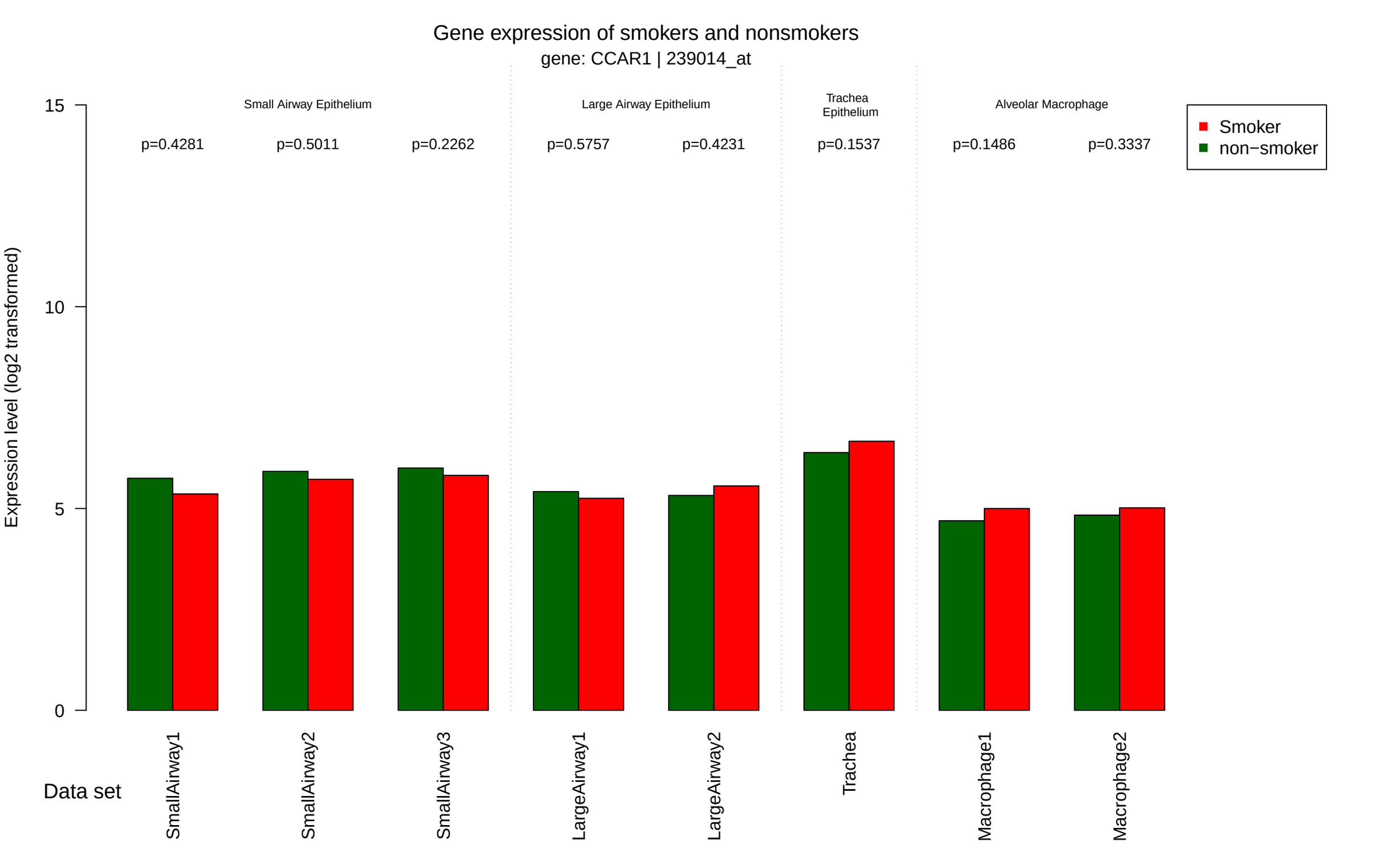
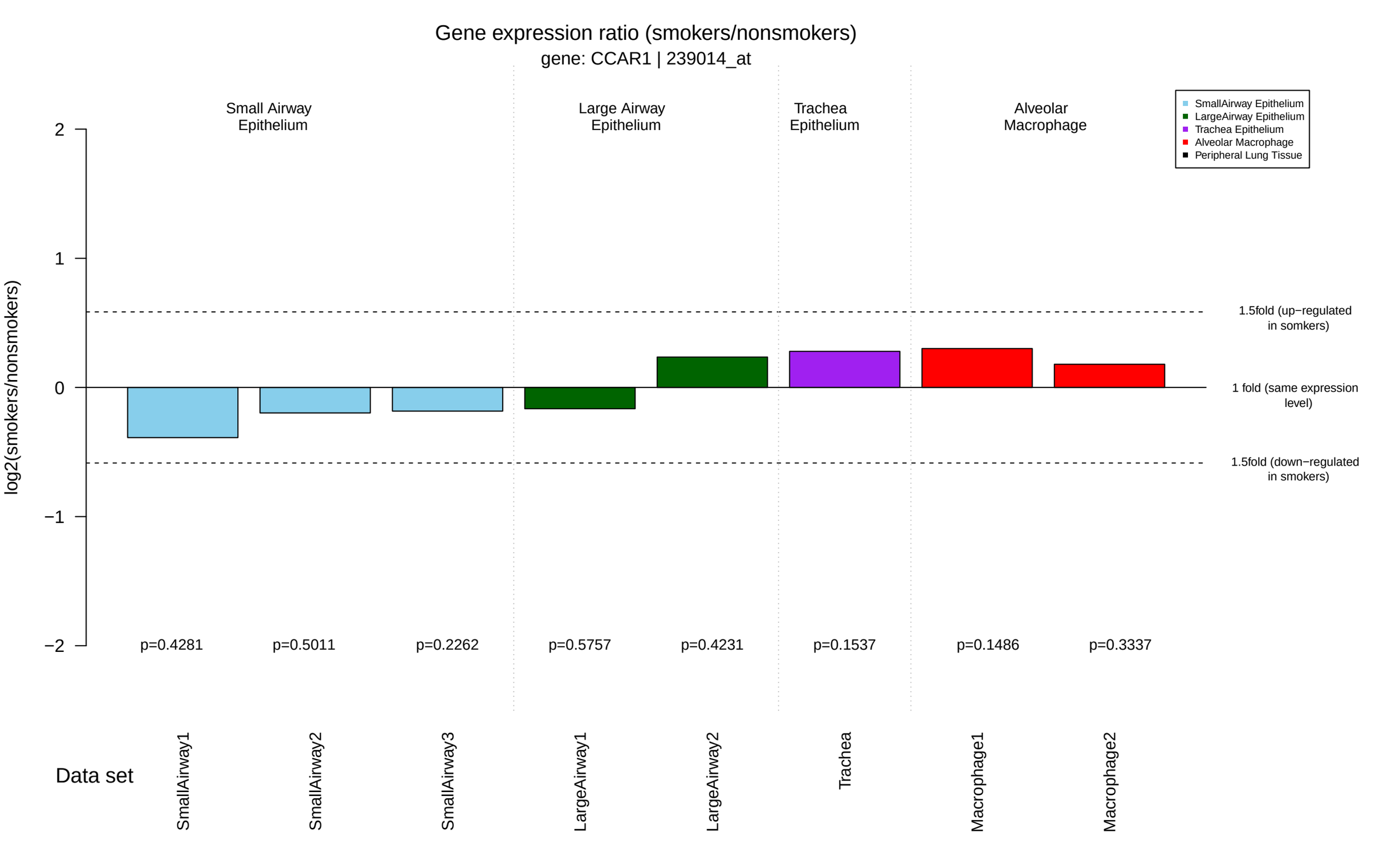
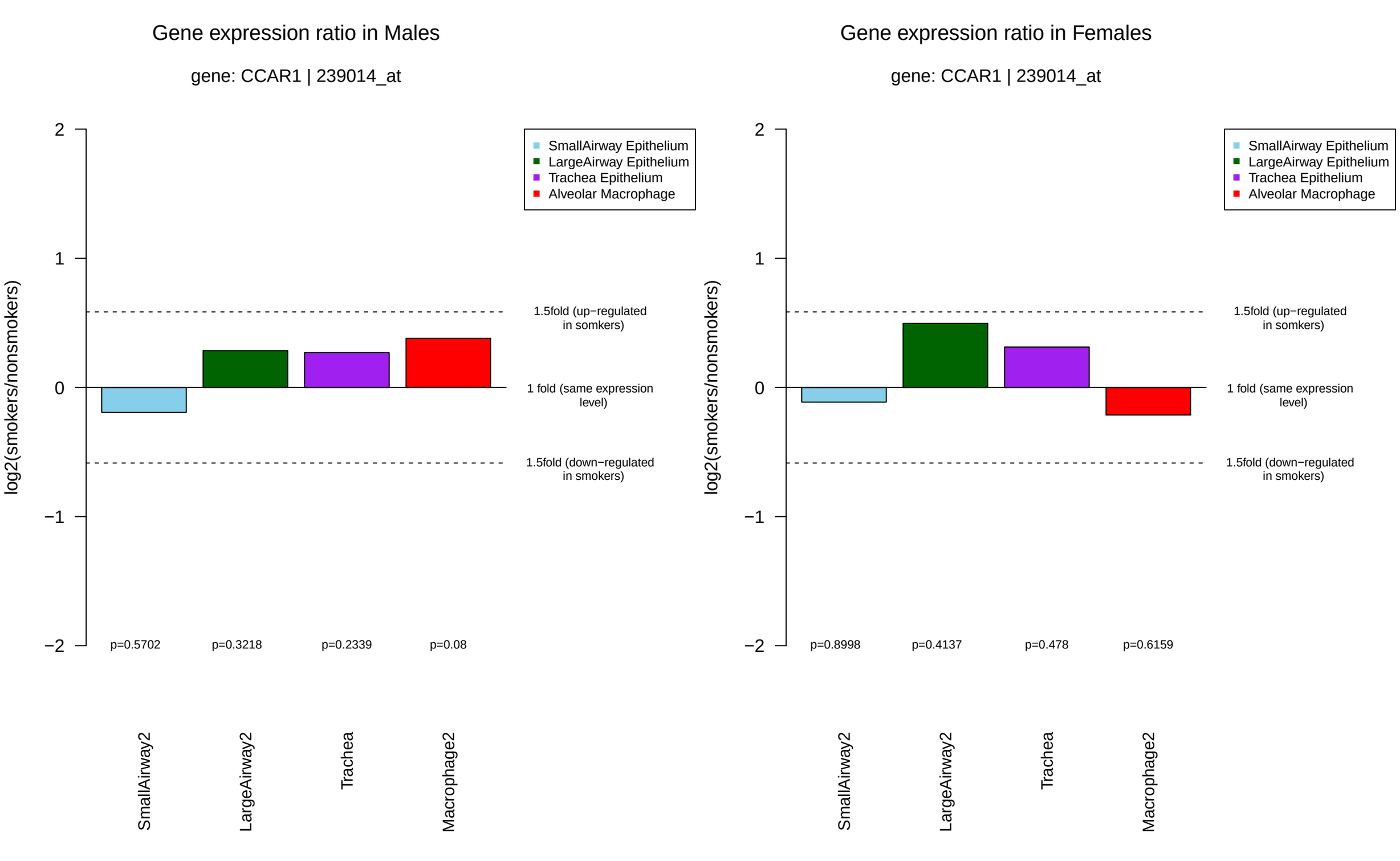
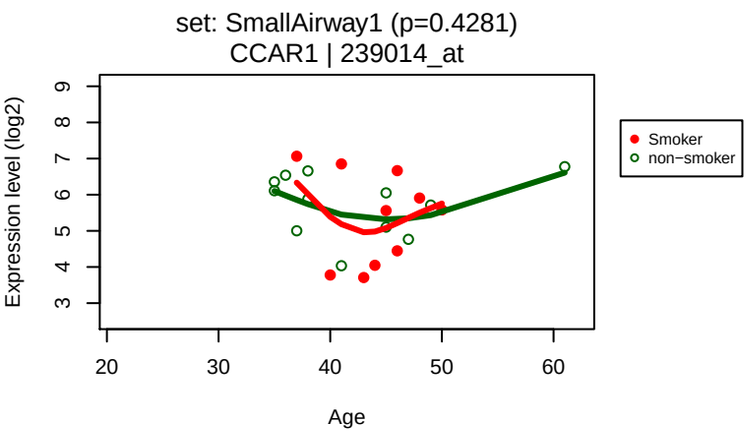
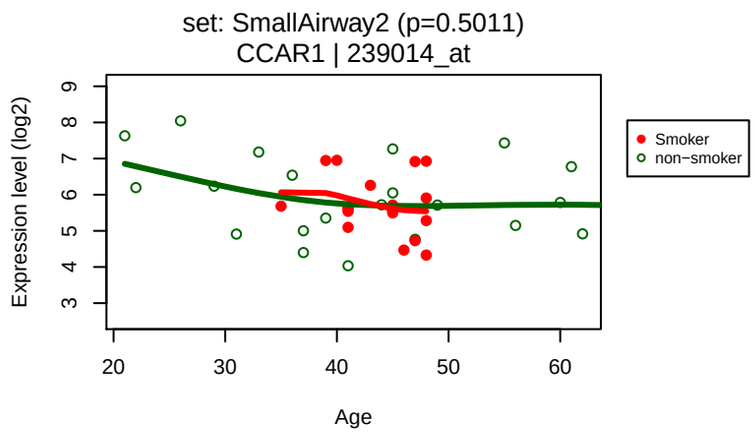
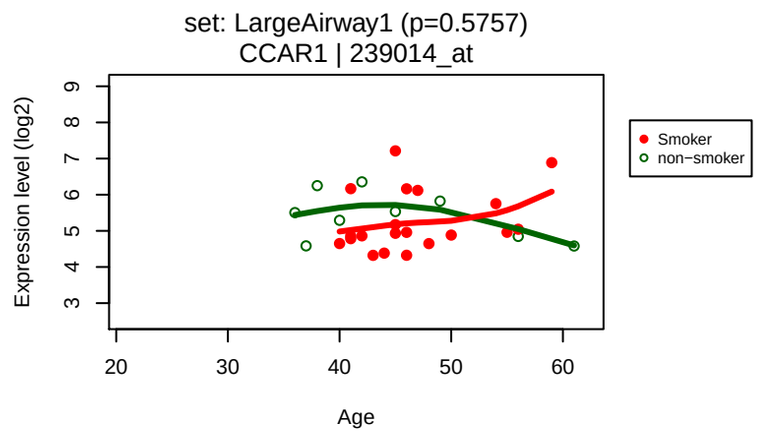

Gene expression of smokers and nonsmokers
gene: CCAR1 | 239014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4281
p=0.5011
p=0.2262
p=0.5757
p=0.4231
p=0.1537
p=0.1486
p=0.3337
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

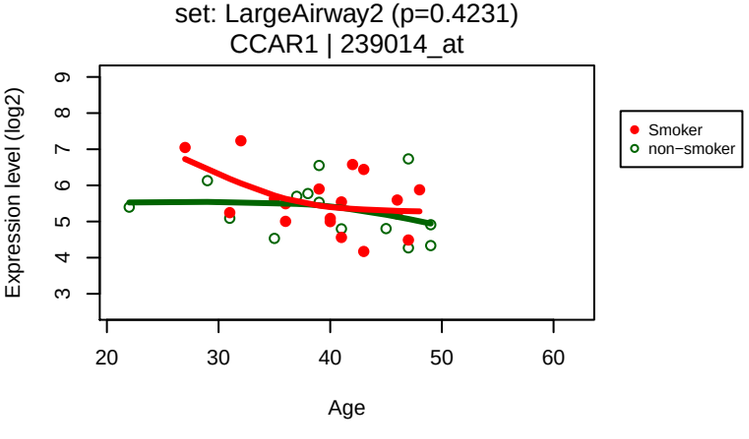

Gene expression ratio (smokers/nonsmokers)
gene: CCAR1 | 239014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4281
p=0.5011
p=0.2262
p=0.5757
p=0.4231
p=0.1537
p=0.1486
p=0.3337
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

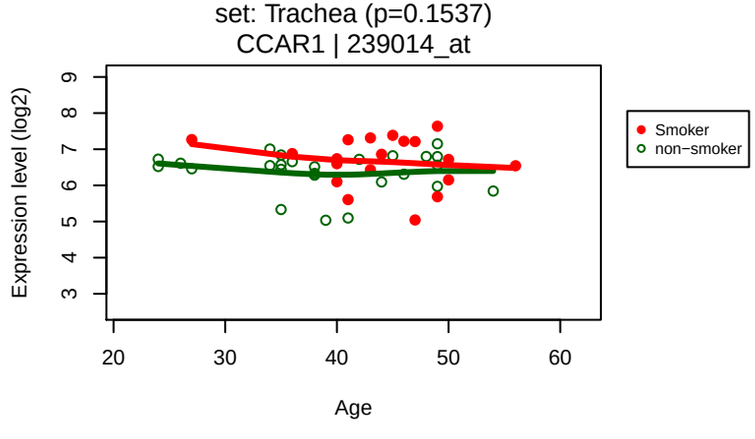

Gene expression ratio in Males
gene: CCAR1 | 239014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5702
p=0.3218
p=0.2339
p=0.08
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CCAR1 | 239014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8998
p=0.4137
p=0.478
p=0.6159
SmallAirway2
LargeAirway2
Trachea
Macrophage2

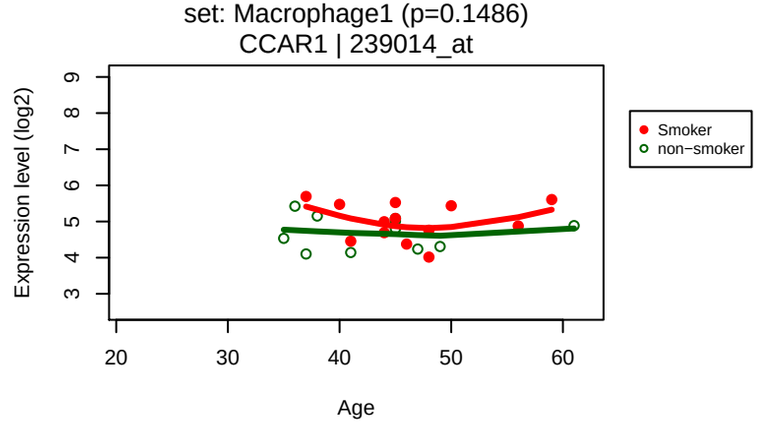

set: SmallAirway1 (p=0.4281)
CCAR1 | 239014_at
Expression level (log2)
Age
Smoker
non−smoker

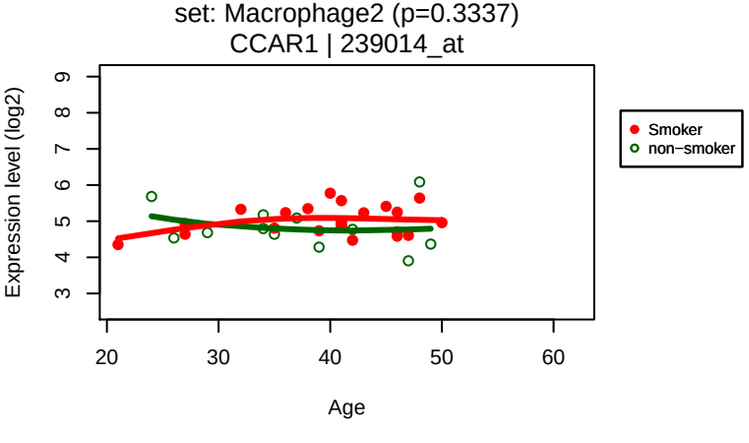

set: SmallAirway2 (p=0.5011)
CCAR1 | 239014_at
Expression level (log2)
Age
Smoker
non−smoker

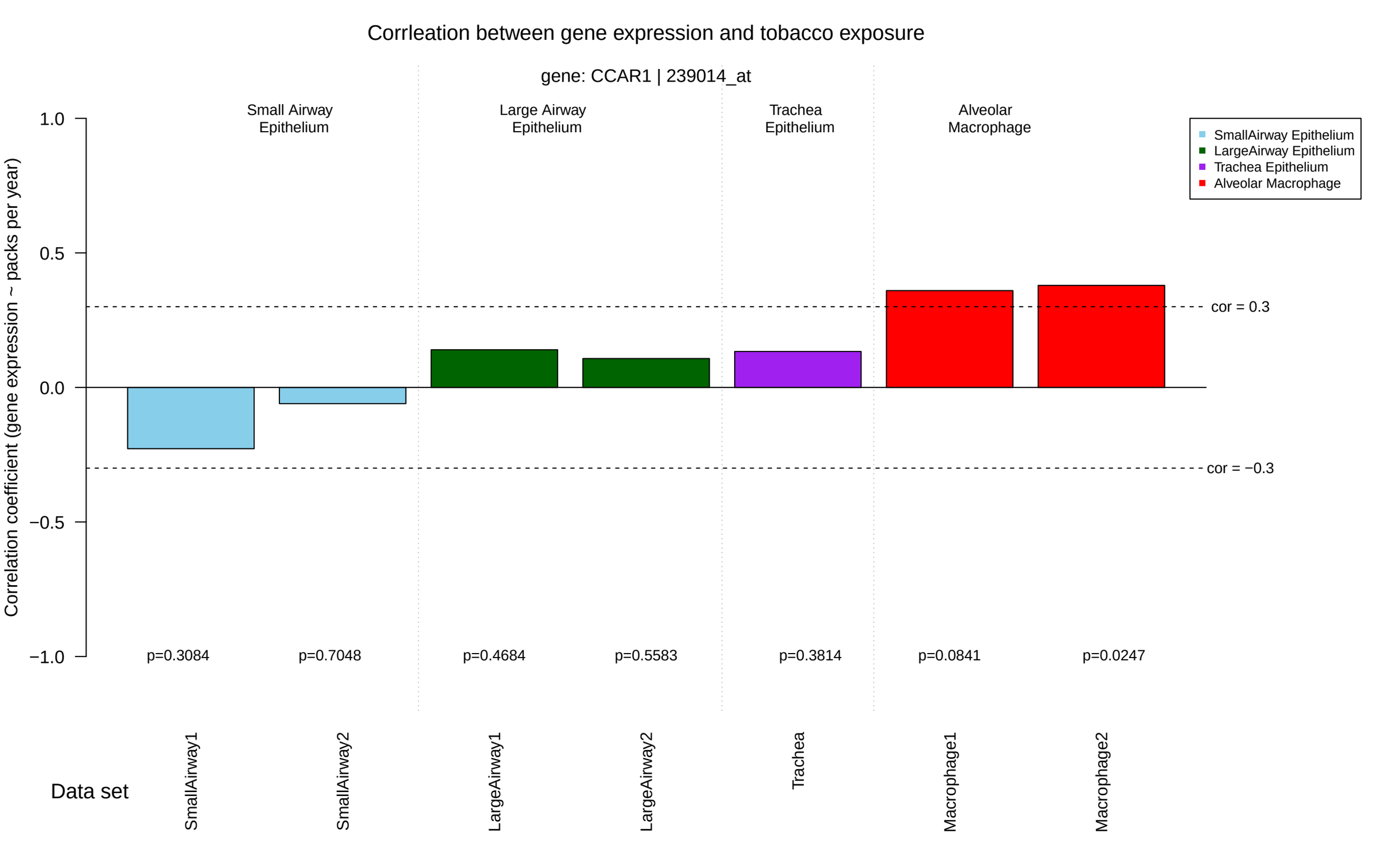

set: LargeAirway1 (p=0.5757)
CCAR1 | 239014_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4231)
CCAR1 | 239014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1537)
CCAR1 | 239014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1486)
CCAR1 | 239014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3337)
CCAR1 | 239014_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CCAR1 | 239014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3084
p=0.7048
p=0.4684
p=0.5583
p=0.3814
p=0.0841
p=0.0247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2